

CSE400 – Lecture 7

Expectation, CDFs, PDFs and Problem Solving

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1. Lecture Structure (Page 2)

According to the outline:

- Cumulative Distribution Function (CDF)
 - Definition
 - Properties
 - Examples
- Probability Density Function (PDF)
 - Definition
 - CDF–PDF Relationship
- Expectation of Random Variables (introduced later)

The lecture follows this order.

2. Intuition: CDF and PDF (Water Tank Analogy)

Given Model

A cylindrical water tank:

- Radius = R
- Height = H
- Water height = h

Interpretation

The slide states:

Height $h \rightarrow$ value of random variable X
Volume up to $h \rightarrow$ cumulative probability

So:

- Each height corresponds to a value of X
- Accumulated water corresponds to $\Pr(X \leq h)$

Thus:

$$\text{Volume up to } h \longleftrightarrow F_X(h)$$

Volume Calculation

Given:

$$V(h) = \int_0^h \pi R^2 dh$$

Step-by-step:

- Cross-sectional area $= \pi R^2$
- Volume $=$ Area $\times$ Height
- Integrate area from 0 to h

Result:

$$V(h) = (\pi R^2)h$$

Probability Interpretation

Maximum volume:

$$V(H) = \pi R^2 H$$

This corresponds to total probability $= 1$.

Thus:

- πR^2 behaves like a constant PDF
- $V(h)$ behaves like a CDF

CDF $=$ accumulated PDF.

3. Cumulative Distribution Function (CDF)

3.1 Definition

The slide defines:

$$F_X(x) = \Pr(X \leq x), \quad -\infty < x < \infty$$

Step-by-step meaning:

- Fix a real number x
- Consider the event: “ $X \leq x$ ”
- Compute its probability
- That probability is $F_X(x)$

So:

The CDF gives the total probability accumulated up to x .

Importance

The slide states:

Most information about the random experiment is determined by the behavior of $F_X(x)$.

Meaning:

Knowing $F_X(x)$ allows us to compute all probabilities.

CDF fully characterizes the random variable.

3.2 Properties of CDF

Property 1: Boundedness

$$0 \leq F_X(x) \leq 1$$

Reasoning:

$F_X(x)$ is a probability.

Property 2: Limits

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

Reasoning:

- At $-\infty$, probability is 0
- At $+\infty$, probability is 1

Property 3: Monotonicity

For $x_1 < x_2$:

$$F_X(x_1) \leq F_X(x_2)$$

Reasoning:

$$\{X \leq x_1\} \subset \{X \leq x_2\}$$

Property 4: Interval Probability

For $x_1 < x_2$:

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

Derivation:

$$\Pr(X \leq x_2) = \Pr(X \leq x_1) + \Pr(x_1 < X \leq x_2)$$

Rearranging:

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

4. CDF Example #1: Validity Check

(a)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$$

Range of $\tan^{-1}(x)$ is $(-\pi/2, \pi/2)$.

So range is $(0, 1)$.

Valid CDF.

(b)

$$F_X(x) = (1 - e^{-x})u(x)$$

Valid CDF.

(c)

$$F_X(x) = e^{-x^2}$$

Invalid.

(d)

$$F_X(x) = x^2 u(x)$$

Invalid.

5. CDF Example #2

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a)

$$\Pr(X > 5) = 1 - F_X(5) = e^{-5}$$

(b)

$$\Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c)

$$\Pr(3 < X < 7) = F_X(7) - F_X(3) = e^{-3} - e^{-7}$$

(d)

Let $A = \{X > 5\}$, $B = \{X < 7\}$.

$$\Pr(A|B) = \frac{F(7) - F(5)}{F(7)}$$

6. Probability Density Function (PDF)

6.1 Definition

$$f_X(x) = \lim_{\varepsilon \rightarrow 0} \frac{\Pr(x < X < x + \varepsilon)}{\varepsilon}$$

6.2 Using CDF

$$\Pr(x < X < x + \varepsilon) = F_X(x + \varepsilon) - F_X(x)$$

$$f_X(x) = \lim_{\varepsilon \rightarrow 0} \frac{F_X(x + \varepsilon) - F_X(x)}{\varepsilon}$$

6.3 PDF–CDF Relationship

$$f_X(x) = \frac{dF_X(x)}{dx}$$

6.4 Inverse Relationship

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

7. Expectation of Random Variables

Only heading is visible.

No further content available.

Final Exam-Oriented Logical Map

CDF

$$F_X(x) = \Pr(X \leq x)$$

Properties:

- $0 \leq F \leq 1$
- $F(-\infty) = 0, F(\infty) = 1$
- Non-decreasing
- $\Pr(a < X \leq b) = F(b) - F(a)$

PDF

$$f(x) = \lim_{\varepsilon \rightarrow 0} \frac{\Pr(x < X < x + \varepsilon)}{\varepsilon}$$

$$f(x) = \frac{dF(x)}{dx}$$

$$F(x) = \int_{-\infty}^x f(t) dt$$